

MMU753
TH#4
YUSUF AĞRI ÖĞÜT
N25240126

Part-3

Q-17)

Following city driving cycle is created using Claude (Reference: UN ECE Regulation 83 / EU Directive 70/220/EEC).



Q-18)

Following efficiency map is created using claudes. Used data for breakpoints and efficiency matrix is as following:

RPM_bp = [800, 1200, 1600, 2000, 2500, 3000, 3500, 4000, 4500, 5000, 5500, 6000]; % [RPM]

Torque_bp = [0, 20, 40, 60, 80, 100, 120, 140, 160, 174]; % [Nm]

Efficiency_torque = [

% 800 1200 1600 2000 2500 3000 3500 4000 4500 5000 5500 6000

3.0, 4.0, 5.0, 5.5, 6.0, 6.0, 5.5, 5.0, 4.5, 4.0, 3.5, 3.0; % T = 0 Nm (idle/friction)

8.0, 11.0, 14.0, 16.0, 17.0, 17.5, 17.0, 16.0, 15.0, 13.5, 11.5, 9.0; % T = 20 Nm

13.0, 18.0, 22.0, 25.0, 27.0, 28.0, 27.5, 26.5, 25.0, 22.5, 19.5, 15.0; % T = 40 Nm

17.0, 22.0, 27.0, 30.0, 32.0, 33.0, 32.5, 31.5, 30.0, 27.5, 24.0, 19.0; % T = 60 Nm

20.0, 26.0, 30.5, 33.5, 35.0, 35.5, 35.0, 34.0, 32.5, 30.0, 26.5, 21.5; % T = 80 Nm

22.5, 28.5, 32.5, 35.0, 36.5, 37.0, 36.5, 35.5, 34.0, 31.5, 28.0, 23.0; % T = 100 Nm

24.0, 30.0, 33.5, 35.5, 37.0, 37.5, 37.0, 36.0, 34.5, 32.0, 28.5, 23.5; % T = 120 Nm

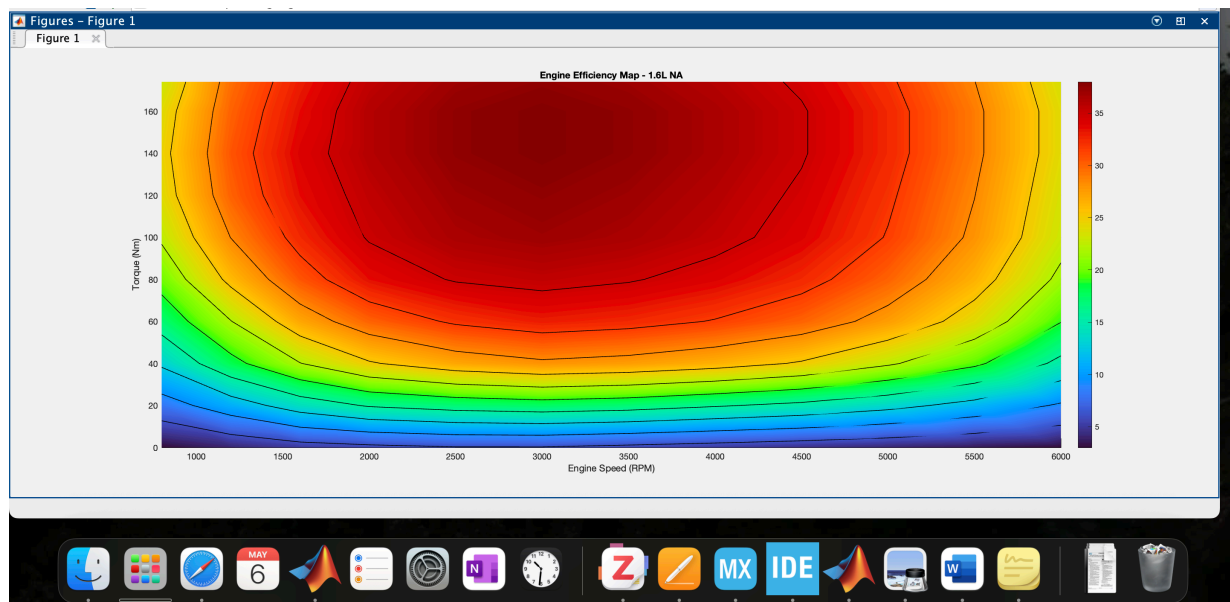
24.5, 30.5, 34.0, 36.0, 37.5, 38.0, 37.5, 36.5, 35.0, 32.5, 29.0, 24.0; % T = 140 Nm

24.0, 30.0, 33.5, 36.0, 37.5, 38.0, 37.5, 36.5, 35.0, 32.5, 29.0, 24.0; % T = 160 Nm

23.0, 29.5, 33.0, 35.5, 37.0, 37.5, 37.0, 36.0, 34.5, 32.0, 28.5, 23.5; % T = 174 Nm (peak

torque)

];



Generator is modeled as 0.9 efficiency.

Motor efficiency isn't modeled due to time concerns.

Q-19)

Acceleration sizing is done using code given in class.

$$P_t = \frac{\gamma m}{2t_a} (V_f^2 + V_b^2) + \frac{2}{3} m g f V_f + \frac{1}{5} \rho S C_x V_f^3$$

- m : mass of vehicle; γ : correction factor for effective mass
- t_a : time to accelerate
- V_b : speed corresponding to switch from constant torque mode to constant power mode

=== Traction Motor Power Rating ===

Inertia term : 58.42 kW

Rolling term : 3.56 kW

Aero term : 3.15 kW

Total P_t : 65.12 kW

Engine sizing is done using code given in class.

$$P_{e/g} = \frac{V}{\eta_t \eta_m} \left(m g f + \frac{1}{2} \rho S C_x V^2 \right)$$

=== Cruise Power Rating at 130 km/h ===

Rolling resistance : 192.1 N

Aerodynamic drag : 479.2 N

Total road load : 671.3 N

Required power P : 33.67 kW

Battery and PPS sizing is done and following is the result:

=== Constraint 1: Energy ===

Target EV range : 20 km

Average power demand : 3.5 kW

Energy at wheel : 3.50 kWh
Battery size (energy) : 9.51 kWh

=== Constraint 2: Power ===

Motor peak power : 69.8 kW
Max C-rate discharge : 20 C
Battery size (motor) : 3.49 kWh
Regen peak power : 30.0 kW
Max C-rate charge : 15 C
Battery size (regen) : 2.00 kWh
Battery size (power) : 3.49 kWh

=== Final Battery Size ===

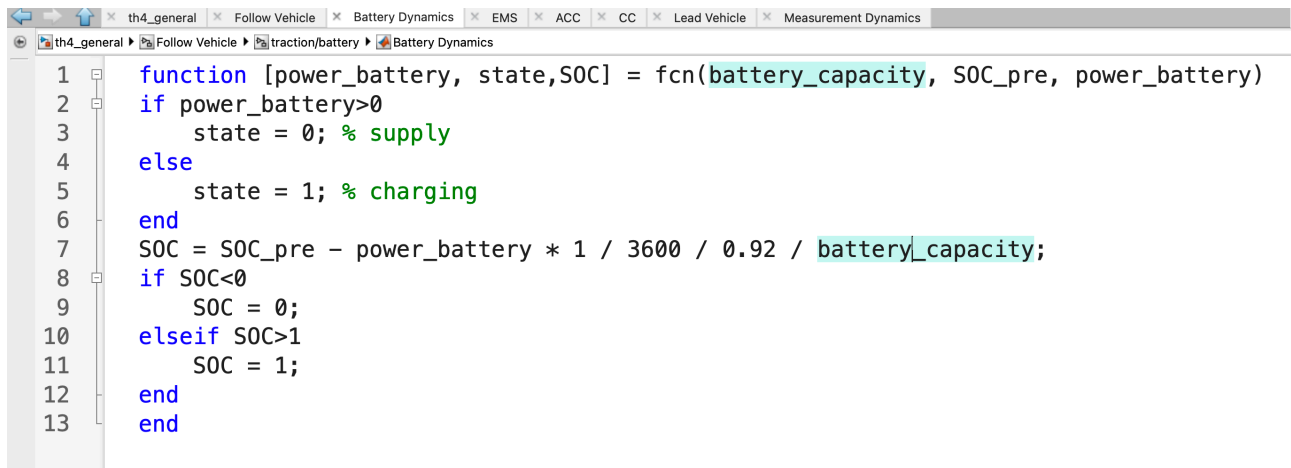
Energy constraint : 9.51 kWh
Power constraint : 3.49 kWh
Driving constraint : Energy (EV range)
Final battery size : 9.51 kWh

=== Derived Battery Parameters ===

Nominal voltage : 300 V
Capacity : 31.7 Ah
Peak discharge current: 233 A
Usable energy : 3.80 kWh
SOC window : 40% - 80%

Q-20)

SOC is modelled in simulink.



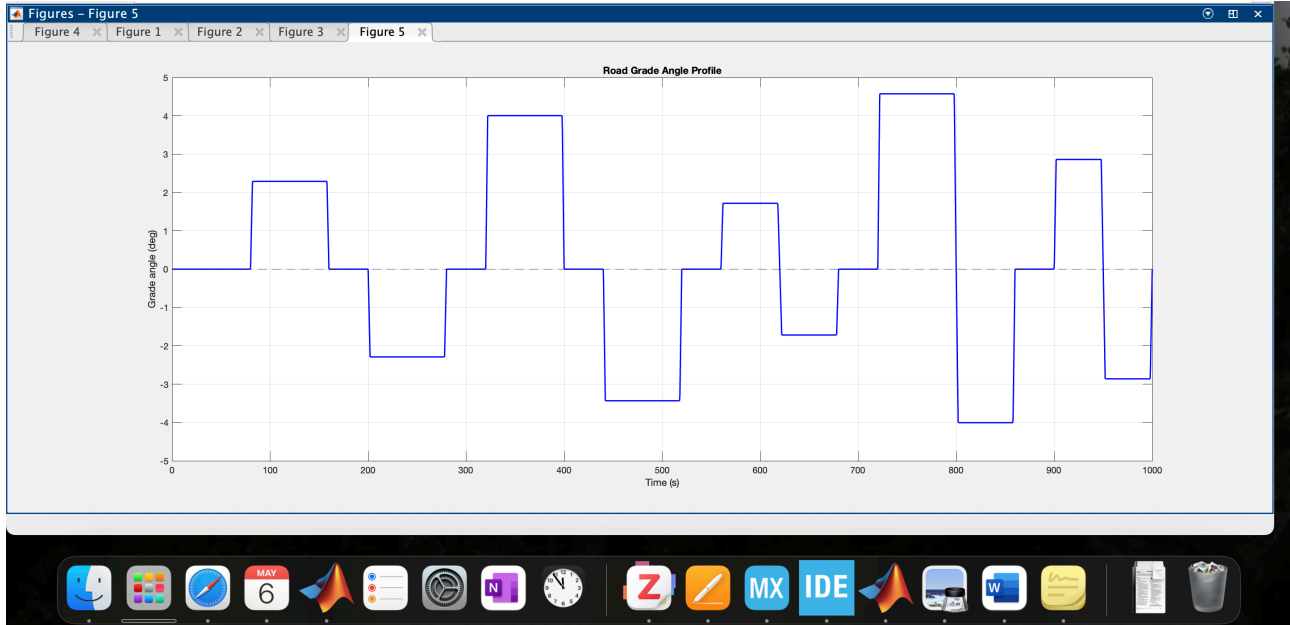
The image shows a screenshot of a MATLAB Simulink script editor. The top window bar displays several open files: 'th4_general', 'Follow Vehicle', 'Battery Dynamics', 'EMS', 'ACC', 'CC', 'Lead Vehicle', and 'Measurement Dynamics'. The 'Battery Dynamics' window is active, showing a function definition for 'fcn'. The function takes three inputs: 'battery_capacity', 'SOC_pre', and 'power_battery'. The function logic is as follows: if 'power_battery' is greater than 0, the state is set to 0 (supply); otherwise, the state is set to 1 (charging). The State of Charge (SOC) is then updated based on the power and capacity. Finally, the SOC is clamped between 0 and 1. The code is as follows:

```
1 function [power_battery, state, SOC] = fcn(battery_capacity, SOC_pre, power_battery)
2 if power_battery > 0
3     state = 0; % supply
4 else
5     state = 1; % charging
6 end
7 SOC = SOC_pre - power_battery * 1 / 3600 / 0.92 / battery_capacity;
8 if SOC < 0
9     SOC = 0;
10 elseif SOC > 1
11     SOC = 1;
12 end
13 end
```

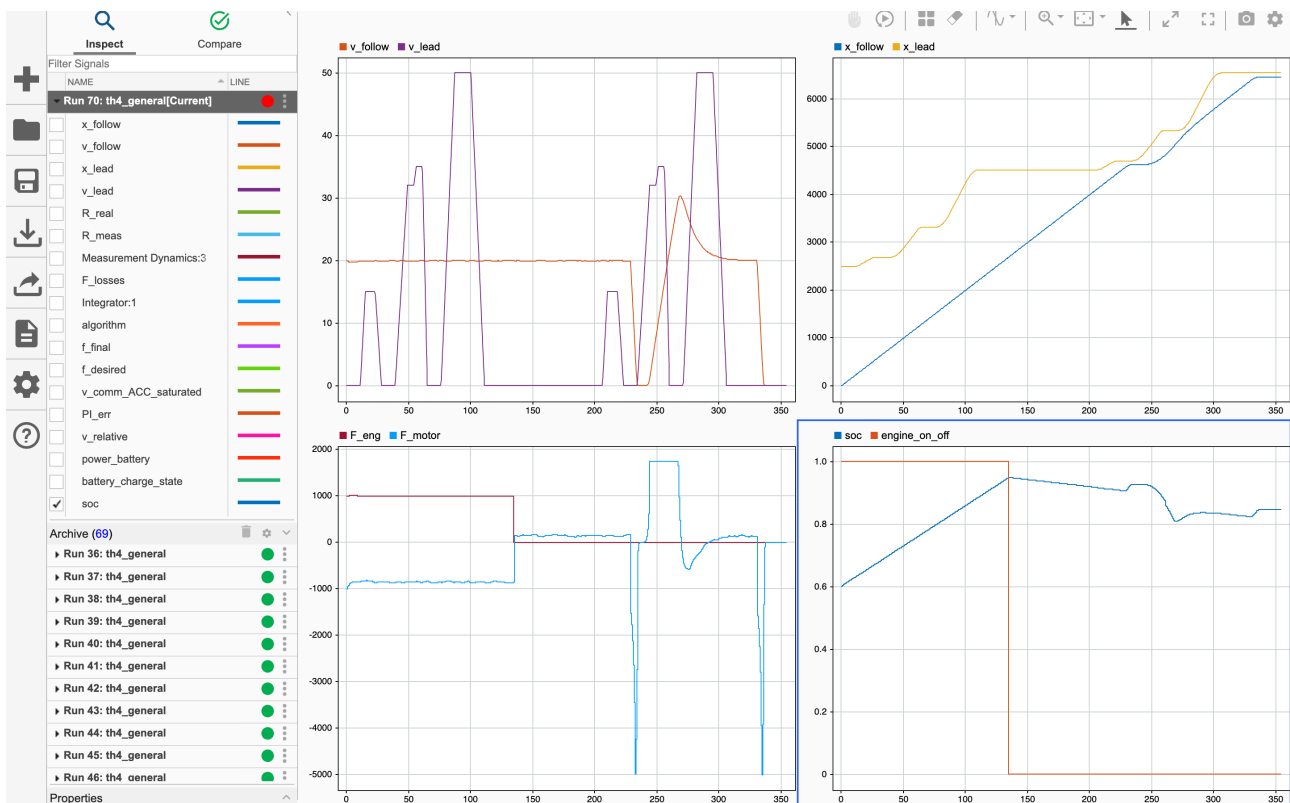
Q-21)

Following conditions are used for simulations:

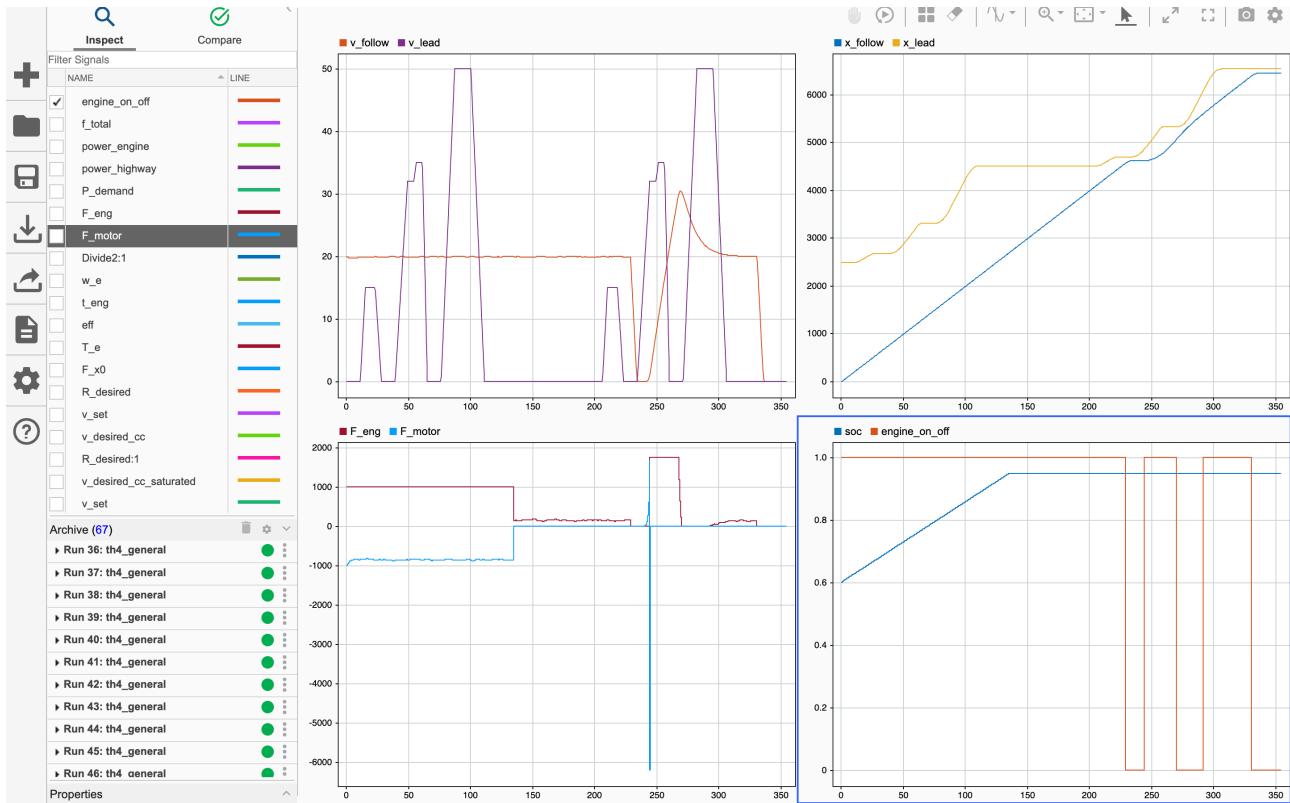
- Follow car has 20 m/s set velocity for CC mode
- Lead car is moving with velocity profile given at the beginning
- For ACC, desired range is 150m
 - Note that this is not very realistic, but tuning couldn't be done in closer distances.
- For seeing charging state changes better, road gradient is added as following:



Following are there results for Engine-on-off and Max SOC algorithms:



Engine on-off control



Max SOC control

It can be seen that in Max SOC strategy, battery charge is tried to be kept at always maximum (%95), and it's successful.